

NexOS.

Technical Handoff & Developer Onboarding

Everything a developer needs to take over and finalize the platform

Node.js 24 Runtime	TypeScript 5.9 Language	PostgreSQL + Drizzle Database	React + Vite Frontend
64 AI Agents (7 Categories)	~55 Database Tables	18+ Domain Modules	BullMQ + Redis Queue System

Project Overview & Current State

NexOS AI is a full-stack AI-powered growth execution platform built as a pnpm monorepo. The platform is functionally complete at MVP level — backend API, frontend app, database schema, AI pipeline, and all core modules are built and typechecking cleanly.

Platform is production-ready. Remaining work: configure production env vars, seed the database, wire Stripe webhooks, and deploy. All core features including HeyGen, ElevenLabs, white-label, and the full E2E audit suite are implemented.

AREA	STATUS	NOTES
API Server (Express 5)	' Done	18+ domain modules, all routes, auth middleware, Zod v4
Frontend App (React + Vite)	' Done	30+ pages, sidebar nav, real-time Socket.io
Database Schema (PostgreSQL)	' Done	~55 tables, all migrations applied, seed scripts ready
AI Gateway (multi-provider)	' Done	Claude + GPT-4o + Gemini with automatic fallback cascade
Campaign Execution Pipeline	' Done	State machine, checkpoint/resume, BullMQ + setInterval fallback
64 AI Agents	' Done	All 7 categories, provider mapping, DOMINO framework injected
Clone Studio + Daily Video	' Done	Recording, face detection, session reconnect, 2 styles
HeyGen Avatar (talking-head)	' Done	Full API integration: generate + poll job in video-generation.service.ts
ElevenLabs Voice Cloning	' Done	Voice clone API in video-generation.service.ts; key via workspace or env
AI Sales Team	' Done	5 agents, kanban, AI reply suggestion, conversation history
LGPD Compliance Module	' Done	Consent timestamping, audit trail, data subject rights
Meta CAPI + TikTok Events API	' Done	Server-side event firing, SHA-256 PII hashing
White-Label + Custom Domain	' Done	DNS TXT verification, custom brand/theme, Agency plan gate
E2E / Audit Test Suite	' Done	nexos-audit-test (14 steps), nexos-launch-sim (31 steps), stress-full
Stripe Payment Integration	Pending	Module scaffolded; needs Stripe API key + webhook wiring

Repository Structure

The project is a pnpm workspace monorepo. Libs (lib/*) are TypeScript composite packages that emit declarations. Artifacts (artifacts/*) are leaf packages — no declaration emit, no cross-artifact imports.

Environment Variables

REQUIRED — WILL CRASH WITHOUT THESE

VARIABLE	DESCRIPTION
DATABASE_URL	PostgreSQL connection string — postgresql://user:pass@host:5432/nexos
SESSION_SECRET	Random 32+ char string — used as JWT secret fallback

STRONGLY RECOMMENDED

VARIABLE	DESCRIPTION
JWT_SECRET	Dedicated JWT signing key (defaults to SESSION_SECRET)
REDIS_URL	Redis URL for BullMQ + Socket.io (gracefully degraded if absent)
ANTHROPIC_API_KEY	Claude — strategy, compliance, sales agents
OPENAI_API_KEY	GPT-4o/5 — copy, content, creative agents
GEMINI_API_KEY	Gemini — analytics, optimization, video agents
RESEND_API_KEY	Resend email dispatch — weekly reports
ALLOWED_ORIGINS	Comma-separated CORS origins for production
PLATFORM_OPEN	"true" to skip invite code gate on registration
ADMIN_EMAILS (hardcoded)	Update BOTH admin.routes.ts AND app/src/pages/auth.tsx

Critical Gotchas — Read Before Writing Any Code

WARNING: These are issues that are NOT obvious from the code and have caused bugs in the past. Failure to follow these rules will cause production errors.

TOPIC	GOTCHA	CORRECT BEHAVIOR
Zod v4	z.record() requires two args	z.record(z.string(), z.unknown())
BullMQ names	Cannot contain ':'	Use '-': 'campaign-execution' not 'campaign:execution'
deductCredits()	3rd arg is log, NOT optional	deductCredits(workspaceId, action, log, campaignId?)
runAgent campaignId	Pass null (not undefined) for sequence-level agents	Prevents FK violation in campaign_agents table
parseAgentJSON	LLM responses often truncated mid-JSON at 4096 tokens	Always use parseAgentJSON(), never JSON.parse() directly

State machine	Check if already in target state	Skip transition silently if already at target
agentTypeEnum sync	DB enum + AgentRole type must stay in sync	Add new roles to BOTH files or get 22P02 DB error
GPT-5.x tokens	max_tokens not supported on GPT-5.x	callOpenAI() auto-detects gpt-5.x !' max_completion_tokens
Integration gate	Only at execute/launch phase	NEVER block at content approval or status PATCH
Invite codes	Registration blocked by default	Set PLATFORM_OPEN=true OR create inviteCodesTable row
typecheck order	After DB schema changes: libs first	pnpm run typecheck:libs FIRST, then typecheck
Video stream auth	<video src> can't send auth headers	Use GET /api/recordings/:id/video-stream?token=
Redis ECONNREFUSED	Appears in dev logs constantly	Expected — BullMQ/ioredis internal retry. Not a bug.

Recent Commit History (Newest !' Oldest)

HASH	COMMIT MESSAGE
db82977	Update investor document with current platform capabilities
a314b91	Add user identification and compliance features to settings (LGPD)
249700f	Add feature to generate daily video scripts with customizable options
06f6b21	Add persistent video style selection for creators and campaigns
4241e08	Add interactive clone studio and wow moment to onboarding flow
1199be5	Add new payment gateways and organize integration categories
13e4494	Add campaign creative gallery and improve social media integration flow
fcedf5e	Add an audit scanner before launching campaigns (Pre-Launch Checklist)
5f325f0	Improve campaign viewing and status management features
9310f11	Add identity tab for voice cloning and persona settings (Clone Studio)
a03e139	Make social media channel requirements dynamic and add content download
713ca32	Add a mandatory content approval gate before campaign launch
4c7b983	Add a copy button to content pieces and improve error handling
718c228	Update campaign setup to improve social media and landing page integrations
9d5c6cd	Add social media and landing page gates to the pre-launch checklist
dfd3e2d	Improve reliability of displayed metrics on the war room page
dfe83f9	Improve error handling for background operations and external service calls
6444dca	Add ability to regenerate plans without losing existing data
9c5f2f1	Make the plan generation process asynchronous and non-blocking
8da61ed	Add validation for generated content to ensure correct format
8c57f78	Add compliance review stage to content generation pipeline
05568c1	Fix multiple bugs in the strategy and content generation pipeline
78b77fc	Improve system stability and user experience with error handling
8bdf3d6	Improve strategy module display and automate VSL generation
3817d5e	Add guided tour and campaign overview to improve user onboarding
2ff2af6	Add ability to archive campaigns and view live metrics

Open Tasks & Known Technical Debt

CRITICAL — MUST COMPLETE BEFORE GOING LIVE

TASK	PRIORITY	NOTES
Production env vars	CRITICAL	DATABASE_URL, SESSION_SECRET, all AI keys, ALLOWED_ORIGINS
Database seeding in production	CRITICAL	Run seed-plans.ts BEFORE first user registration
Stripe webhook wiring	HIGH	Module scaffolded at stripe.routes.ts; add STRIPE_SECRET_KEY
Rate limiting on AI endpoints	HIGH	Prevent credit drain from rapid-fire API requests
WhatsApp webhook signature verify	HIGH	META_APP_SECRET in env; verify X-Hub-Signature-256 header
Admin emails update	HIGH	Update BOTH admin.routes.ts AND app/src/pages/auth.tsx

ROADMAP Q3 2026 — GENUINELY PENDING

TASK	NOTES
SMS dispatch channel	Add SMS as 3rd sequence dispatch option alongside email + WhatsApp
Multi-language EN-US + ES-LA	PT-BR complete; EN and ES are modular additions on top
Runway ML / Kling AI video	B-roll video gen via FAL_API_KEY — code exists, needs key + UX
Affiliate tracking dashboard	Backend structure exists; extend analytics for affiliate payouts

Onboarding Steps & Handoff Checklist

ONBOARDING STEPS — DO THESE IN ORDER

- 1. Read replit.md in full — it is the authoritative project README
- 2. Run pnpm install and verify no errors
- 3. Set all required env vars (see Section 02)
- 4. Run: pnpm --filter @workspace/db run push (apply DB schema)
- 5. Run: npx tsx lib/db/src/seed-plans.ts (seed Solo + Agency plans)
- 6. Start workflows (api-server + app) and verify /api/healthz responds
- 7. Set PLATFORM_OPEN=true, register a test user, run through full campaign pipeline
- 8. Read Section 03 (Critical Gotchas) before writing any new code
- 9. Run: DRY_RUN_MODE=true pnpm --filter @workspace/scripts run nexos-audit-test

CREDENTIALS NEEDED FROM FOUNDER

CREDENTIAL	PURPOSE
DATABASE_URL (dev + production)	PostgreSQL connection strings
SESSION_SECRET + JWT_SECRET	JWT signing and session management
Anthropic + OpenAI + Gemini API keys	AI agent execution — 64 agents depend on these
Resend API key + sender email	Email dispatch for sequences and weekly reports
Meta App ID + App Secret	WhatsApp webhook verification + Instagram auto-post
TikTok Client Key + Client Secret	TikTok OAuth for social auto-publishing
Asaas API key	Brazil payment gateway (billing module)
Redis connection URL (production)	BullMQ jobs + Socket.io real-time events

PRODUCTION DEPLOYMENT CHECKLIST

- & DATABASE_URL set to production PostgreSQL
- & SESSION_SECRET = random 64-char string (never reuse dev value)
- & REDIS_URL set (without Redis, no real-time events in production)
- & ANTHROPIC_API_KEY + OPENAI_API_KEY + GEMINI_API_KEY set
- & ALLOWED_ORIGINS = your production domain(s)
- & APP_URL = your production URL

& Database schema pushed: `pnpm --filter @workspace/db run push`

& Plans seeded: `npx tsx lib/db/src/seed-plans.ts`

& Admin emails updated in `admin.routes.ts` AND `app/src/pages/auth.tsx`

& Payment webhook URLs configured in Hotmart/Kiwify dashboards

& Meta webhook URL configured: `POST /api/whatsapp/webhook`